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Image and metadata-driven personality inference for career recommendation: a social media-based AI framework for adolescents

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Abstract

This study presents a novel AI-based framework that leverages Instagram image and metadata analysis to infer Big Five personality traits and deliver personalized career recommendations for high school students in the UAE. Addressing the limitations of traditional recommender systems that rely on self-reported questionnaires or text, the proposed approach uses multimodal visual features—including profile metrics, HSV color patterns, semantic image labels, and texture analysis—to enable a non-intrusive, scalable personalization method. A pilot study involving data from 30 student accounts served as a proof of concept. Correlation analysis identified profile and HSV features as the most predictive, and four machine learning models were trained, with Logistic Regression achieving 97% accuracy (AUC 0.97) in personality prediction. The inferred traits were mapped to academic majors using a stereotype-based recommender system, achieving 90% alignment with student preferences as measured by the Electronic Emirati Scale for Professional Inclinations (EESPI). Findings demonstrate the feasibility and promise of integrating AI-driven image analysis with personality-aware recommendation. This work contributes to emerging trends in non-verbal, visual data-based personalization, particularly in educational and career guidance domains.

Keywords Recommender system, Machine learning, Social media mining, Personality analysis, Image recognition, Big five, Career recommendations

1 Introduction

Selecting the right academic major is a pivotal decision that has far-reaching consequences for a student's career and future opportunities. Several factors influence this choice, including career prospects, job market conditions, personal interests, family expectations, and financial considerations [1]. As a result, career recommendation systems have become an essential component of educational guidance. Traditional career recommendation systems base their suggestions primarily on students' academic performance and skills [2–4], as well as their learning preferences [5]. However, such methods offer limited personalization, especially for high school students whose curricula often



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lack specialization, resulting in generic recommendations that fail to reflect individual differences.

More critically, students at this early developmental stage often lack the self-awareness or maturity needed to accurately assess their own preferences, strengths, and traits. As such, career guidance that depends heavily on self-reported inputs may not be reliable. While recent studies have demonstrated strong correlations between personality traits and preferences—including career and academic choices—most personality-driven recommender systems still rely on explicit, self-report questionnaires to capture personality data [6, 7, 10]. These instruments are not only time-consuming but also subject to response bias, social desirability effects, and incomplete responses, especially in adolescent populations.

To overcome these limitations, recent advancements in user modeling have focused on implicitly inferring personality traits by analyzing behavior on social media platforms. Instagram, a widely popular platform among high school students, provides rich behavioral signals through images, posts, comments, and engagement metrics [11]. Prior research has shown that Instagram profiles [12, 13], user posts [14–16], and image content [17, 18] can serve as reliable indicators of personality traits, enabling passive and scalable data collection without requiring users to explicitly self-report.

In this study, we propose a novel career recommendation framework that infers personality traits from Instagram data, including behavioral and profile metadata, and aligns these traits with the Big Five personality model [19]. The Big Five personality model, is a widely accepted framework with five dimensions extraversion, agreeableness, conscientiousness, neuroticism, and openness. Conscientiousness is associated with organization and goal orientation; neuroticism is associated with emotional instability and vulnerability to negative affect; extraversion is associated with social engagement and energy; agreeableness is associated with interpersonal cooperation and trust; and openness is associated with intellectual curiosity and creativity. The Big Five model is appropriate for educational and career counseling applications because it offers a succinct yet thorough depiction of personality features.

This framework experiments with two sets of Instagram features—profile metrics and posted/liked images—representing a relatively unexplored area in personality-based recommender systems. By identifying which Instagram features most accurately reflect personality, the system generates personalized career recommendations tailored to students' traits. The proposed recommendations are validated against the career preferences of 30 high school students, providing a gold standard for comparison.

This study is guided by the following research objectives:

- RO1: To develop and evaluate machine learning models for accurately predicting the Big Five personality traits of high school students based on their Instagram activity.
- RO2: To analyze and identify which categories of Instagram-derived features—including profile metadata, image content, and low-level visual attributes (e.g., color and texture)—exhibit the strongest associations with the Big Five personality traits.
- RO3: To investigate the effectiveness of leveraging predicted personality traits to generate personalized career recommendations that align with students' self-reported career preferences.

The remainder of this paper is organized as follows: Sect. 2 reviews the literature on personality prediction and recommender systems. Section 3 outlines the implementation of the proposed career recommendation framework. Section 4 presents the experimental results and findings. Finally, Sect. 5 discusses the results and their implications for future research.

2 Literature review

Recently, there has been a surge of research interest in more personality-oriented approaches within recommender systems. Traditionally, recommender systems focus on user preferences, ratings, and behavioral data. However, the personalization of these systems is limited to explicit data. Hence, researchers are now exploring implicit psychological aspects to enhance recommendation quality, including personality. In this section, we review and analyze the literature related to personality prediction as well as recommender systems based on personality traits.

2.1 Personality prediction

Various personality prediction approaches utilize and analyze different types of user data. Some approaches utilize user posts and comments [1, 2], user profiles [3–5], and posted images [6, 7]. For instance, Christian et al. [1] and Zhu's [2] employ multi-level Deep Learning architectures as well as combined data mining and Deep Neural Networks to predict user personality from user posts on social media platforms, respectively. These methods are robust in handling large-scale textual data and can capture complex linguistic patterns indicative of personality traits. However, they require substantial computational resources and extensive labeled data for training.

Other approaches leverage profile pictures for personality prediction [4, 5]. Profile pictures provide valuable emotional and visual cues, such as facial expressions and presentation style, which can be indicative of personality traits, making them a useful tool for personality prediction. For instance, Liu et al. [4] utilized the emotions portrayed in profile pictures, establishing a correlation between visual cues in profile pictures and personality. While these methods can capture visual aspects of personality, they may be limited by the quality and consistency of profile images, which are often influenced by social desirability bias. To address this limitation, other approaches leverage profile metadata including followers, followings, likes, and number of comments for personality prediction. Such profile metrics provide quantifiable data that offer more objective insights into a user's personality compared to subjective interpretations of facial expressions or presentation styles in profile pictures. For instance, Basaran et al. [3] deploys an Artificial Neural Network (ANN) to predict Facebook users' personalities using metadata such as the number of likes, events, groups, tags, updates, network size, relationship status, as well as age and gender. These metrics reflect a user's social behavior, interests, and level of activity, which are all relevant factors in predicting personality traits.

User-posted images often reflect important aspects of personality, making them a popular choice for personality prediction on social media. Image-based personality prediction has seen diverse approaches focusing on different image features including visual [8–10], content [6–8, 10, 11], and facial features [6, 7, 9]. For instance, Xu et al. [6] deployed a deep Neural Network for multi-dimensional personality characteristics prediction using 2.5 facial contour images. Moreover, Ferwerda et al. [12] utilized visual

features such as hue, saturation, and intensity value (HSV) for personality prediction from user-posted images on Instagram. However, the image's visual features alone do not provide sufficient cues to identify personal behaviors that may be helpful in recommender systems. Hence, Saitov et al. [7] analyzed clusters of facial images to associate visual features with personality traits. These image-based approaches highlight that various image features, including visual, content, and facial features, are strongly correlated with personality traits. For personality prediction and associated behavioral inference problems, recent multimodal and deep learning algorithms have been investigated more and more. To illustrate the potential of multimodal feature fusion for better trait estimation, recent research has examined multimodal personality prediction using deep-learning methods that incorporate visual, text, and other modalities to determine Big Five traits from various sources of data [13]. The benefits of structured multimodal learning were demonstrated in a 2025 study that suggested an adaptive graph transformer network to simulate interactions across modalities for personality prediction [14]. In order to enhance trait prediction performance, another 2025 study used multimodal analysis on Instagram by merging metadata, captions, and photographs. This study demonstrated the importance of integrating various social media signals [15].

Comparative Discussion

Prior research on personality prediction from social media has explored text-based methods using deep learning on user posts [1, 2], profile-based approaches leveraging user metadata [3–5], image-based models exploiting visual and facial cues [6–12], and more recently, multimodal deep learning frameworks that fuse text, images, and metadata for improved Big Five trait estimation [13–15]. While text-based and multimodal approaches achieve strong predictive performance, they often require large labeled datasets, high computational resources, and raise language or privacy concerns, whereas image- and metadata-based methods are more lightweight but typically treat personality inference as an end goal rather than a means to support decision-making. In contrast, this work advances the literature by moving beyond personality prediction to a personality-aware career recommendation framework, integrating Instagram metadata and image-derived features to infer personality traits and directly map them to career preferences. Unlike prior studies, the proposed approach emphasizes actionable educational guidance, remains resource-efficient and privacy-conscious, and is empirically validated against the EESPI career preference scale within a regionally grounded educational context, addressing a key gap between personality modeling and real-world career recommendation systems.

2.2 Role of personality in recommender systems

Personality plays a crucial role in shaping individuals' preferences, behaviors, and decision-making processes. Recommender systems leverage personality traits to offer personalized suggestions in various fields including e-commerce, transportation, agriculture, health care, entertainment, and educational applications [16]. Recommender systems provide personalized suggestions for items that are most relevant to users' metadata collected over time. Personality-based recommender systems have proven to be efficient and have attracted significant attention [12]. Users' personality data is leveraged for various recommender systems [17–19]. For instance, Su et al. [18] proposed a personality-based recommender system for courses and related jobs based on

personality similarity and course grades. Despite the efficiency of the proposed method, it's time-consuming and lacks feasibility, limiting its practicality in real-world applications. To address these limitations, social media platforms are increasingly utilized for feasible and more efficient user data acquisition.

Social media platforms offer a rich source of representative user personality, preferences, and behavioral data. Hence, more recent approaches extract user personality data from social media platforms such as Instagram, Facebook, X (formerly Twitter), and Telegram, among others. Analyzing users' posts and posted images provides key insights that aid in predicting users' personalities, which are essential for personalized recommendations. User posts provide key insights into the users' personalities, interests, and preferences, as well as their knowledge levels [20]. In addition, image posts provide visual features that can be utilized for customized recommendations. For instance, Nalmpantis [19] proposed a movie recommendation system from visual features, which has proven to be more efficient than non-visual features.

Personality-based recommender systems have shown superior efficiency compared to other approaches. Traditional recommender systems often rely on explicit user inputs, such as ratings and preferences, which can be limited in scope and prone to biases. In contrast, personality-based systems leverage more implicit and holistic indicators of user behavior, such as social media posts and images. While this approach offers significant advantages in terms of personalization and accuracy, it also presents challenges related to privacy, data accuracy, and the dynamic nature of social media content. Addressing these challenges is crucial for ensuring the effectiveness and reliability of personality-based recommender systems.

2.3 Career recommendation

Despite the widespread use of personality in recommender systems across various fields, its application in career recommendation remains limited. Career recommendation systems traditionally focus on skills and qualifications [21–24], as well as job preferences [25, 26]. Although profile and skill-based recommender systems provide recommendations that highly match the students' skill sets, high school students do not often have sufficient experience and well-developed skills, which hinders the applicability of such systems. Moreover, such approaches overlook the significant impact of personality traits on career satisfaction and success. Integrating personality-based recommendations into career guidance can provide more holistic and personalized career advice, aligning individuals' personalities with suitable career paths.

In addition, the majority of the preference-based recommendations use explicit questionnaires, which are prone to bias and inaccurate responses due to their reliance on users' self-assessments. Therefore, some methods used varying combinations of user data to provide personalized recommendations, most commonly, skills, preferences, and career goals [23]. However, none of these methods consider the student's personality, which is a key determinant in making informed career choices. Hence, a few approaches provided personality-based recommendations [27, 28]. Despite the efficiency of these approaches, such systems were limited in terms of career options coverage. For instance, Ngiam et al. [27] and Siswipraptini et al. [28] proposed recommender systems that provide personalized career recommendations within the Information Technology field.

The limited use of personality in career recommendation systems represents a gap in the literature and a potential area for future research. By incorporating personality traits into career recommendation models, it is possible to enhance the accuracy and relevance of career advice, ultimately helping individuals make more informed and fulfilling career choices. Furthermore, despite the advancements in personality prediction, there is a noticeable gap in leveraging implicit features for personality prediction from social media content to provide personalized career recommendations.

3 The proposed personalized career recommendation framework

This study proposes a personalized career recommendation system that infers students' personalities from their social media profiles, specifically Instagram. The approach utilizes Instagram profile features and image data to predict personality traits based on the Big Five personality model. The framework consists of three core modules: data acquisition, personality prediction, and personalized recommendation.

The first module involves gathering social media data from Instagram profiles, including posted and liked images, along with ground truth data collected through two questionnaires: the Big Five personality test and the Electronic Emirati Scale for Professional Inclinations (EESPI) career test.

The second module focuses on selecting the most representative Instagram features for personality prediction by calculating correlations between Instagram data and the Big Five personality traits. These selected features are then employed to train several machine learning (ML) models for personality prediction.

In the final module, the predicted personality traits serve as inputs to a recommendation engine that generates personalized career suggestions based on personality profiles.

For validation, the system compares career preferences obtained from the EESPI test with the recommendations generated by the proposed method, evaluating its accuracy and effectiveness (Fig. 1).

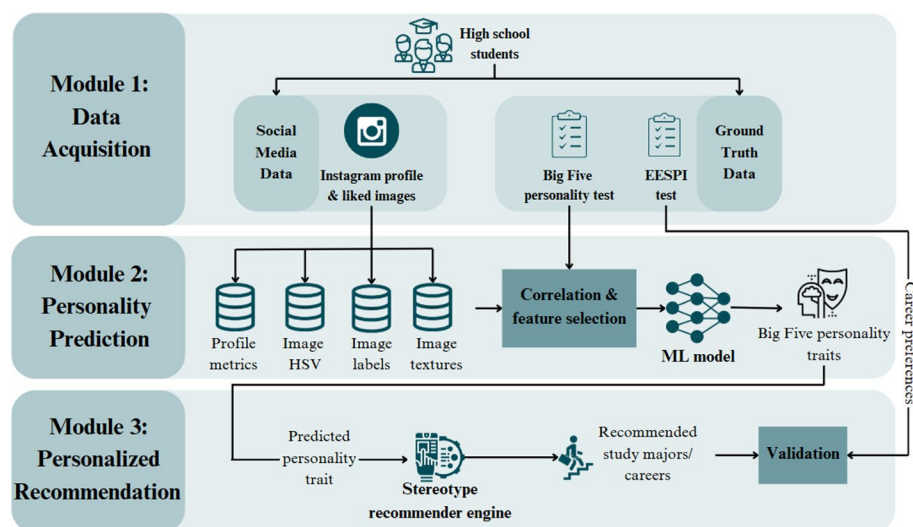


Fig. 1 The proposed personalized career recommendation framework

3.1 Module 1: data acquisition and preprocessing

3.1.1 Participants and recruitment

This study involved a sample of 30 high school students from the United Arab Emirates. All participants were 18 years old or above and the participation was voluntary, and students took part in the study after agreeing to complete the required psychometric assessments and provide access to their Instagram data. All participants completed the Big Five personality questionnaire and the Electronic Emirati Scale for Professional Inclinations (EESPI) test prior to social media data collection.

Participation required that students were active Instagram users with sufficient posted or liked content to enable feature extraction. Inactive accounts and accounts exhibiting bot-like behavior were excluded from the study to ensure data validity and reliability. All included accounts corresponded to human-operated profiles that demonstrated regular engagement activity.

3.1.2 Social media data acquisition

This study uses Instagram profiles of high school students to infer personality traits for personalized career recommendations. Four distinct datasets are constructed: Instagram profile metrics, image HSV (Hue, Saturation, and Value), image labels, and image textures. These feature modalities are extracted from both posted and liked images, and each is analyzed to determine its predictive relationship with the Big Five personality traits. The subsequent sections provide a detailed explanation of each dataset, accompanied by illustrative examples.

Instagram Profile Metrics

User profile metrics encompass various aspects of a user's online activity and preferences, offering insights into their behaviors, tendencies, and personality traits [3–5]. For instance, a large social network can indicate extraversion, while smaller networks may indicate introversion. In addition, metrics such as the frequency of posts, likes, or comments can indicate traits like extraversion and introversion. This work utilizes varying profile metrics, including the total number of followers, followings, posts, likes, and comments. Table 1 demonstrates a sample of the profile metrics for five random students, each exhibiting dominant traits in one of the Big Five dimensions. The inclusion of diverse user profiles illustrates how various patterns of activity can correspond to different personality characteristics. As depicted, the extraverted user has the highest social engagement metrics, while the neurotic user has the lowest engagement.

The same users featured in Table 1 are carried through the subsequent sections and analyses, allowing for a consistent examination of how various Instagram-derived features correspond to personality traits.

Table 1 Sample Instagram profile metrics features

Profiles metric feature	User 1	User 3	User 8	User 12	User 13
	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
Followers Count	<u>2107</u>	1955	837	46	46
Followings Count	<u>1034</u>	460	177	26	111
Posts Count	<u>184</u>	21	48	20	21
Likes Count	<u>4070</u>	1046	1759	10	90
Comments Count	<u>237</u>	23	22	0	17

Bold and underlined indicate the highest values, while **bold** indicates the lowest values

For the posted and liked images, the top 20 images for each student are crawled, resulting in a representative sample of their visual content. These images are then used as the basis for extracting visual features, including color composition, semantic content, and texture patterns. Figure 2 depicts a random sample of these images taken from the students' Instagram profiles. Notably, the visual characteristics of the images vary among the different personality traits.

Image HSV .

Image colors reflect emotions and moods, which are closely associated with psychological and personality traits [4, 29]. For instance, warm colors are associated with energy and extraversion, while cool colors are associated with calmness and introversion, as depicted in Fig. 2(a) and (d).

Unlike the RGB color model, the HSV color model is a representation of color that aligns closely with human visual perception. The HSV color model utilizes three distinct components: Hue, Saturation, and Value [30]. Hue specifies the color category, such as red, green, or blue. Saturation quantifies the intensity or purity of the color, with higher saturation values indicating more pure colors, while lower values correspond to desaturated tones. Value denotes the brightness or luminance of the color.

In this work, the Hue, Saturation, and Value features are extracted for each image. For a comprehensive color representation analysis, this work utilizes the primary and secondary hue values, the minimum, maximum, mean, standard deviation, and median of the saturation, as well as the intensity value. For each user, the HSV components are computed and aggregated across their 20 selected images, providing a user-level summary that reflects their overall visual style. Table 2 depicts the mean and standard deviation of HSV features of all retrieved images for the same 5 sample users. Each user in Table 2 is associated with a dominant personality trait from the Big Five, allowing for exploring patterns and underlying personality dimensions.

Image Labels .

The semantic content of images provides another layer of psychological insight. Personality traits significantly influence individual preferences, which are often expressed through the selection and sharing of images on social media platforms [7]. Image labels and descriptions provide valuable insights into these traits by capturing the content and stylistic elements of the images [31, 32]. For instance, a preference for nature-related labels could indicate agreeableness and calmness [31]. Similarly, an interest in food and drink might suggest extroversion, as depicted in Fig. 2(a). Meanwhile, labels reflecting art, fashion, or design may be associated with creativity and openness, as depicted in Fig. 2(e).

Hence, this work utilizes image labels as a predictive feature of personality. To extract descriptive labels, image recognition is performed using Google Vision API, which is



Fig. 2 Sample Instagram posts and liked images

Table 2 Instagram image HSV features (Mean $\pm$ Std) for 5 sample users

Image feature		User 1	User 3	User 8	User 12	User 13
		Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
Hue	Red	253265.55 $\pm$ 196320.06	152608.65 $\pm$ 143410.12	282720.75 $\pm$ 200156.30	78482.85 $\pm$ 79258.04	165108.9 $\pm$ 214839.76
	Green	48301.3 $\pm$ 54453.70	172321.15 $\pm$ 244419.83	196820.3 $\pm$ 217743.55	436835.65 $\pm$ 341457.02	228159.3 $\pm$ 263421.67
	Blue	60376.85 $\pm$ 106075.73	46399.35 $\pm$ 64748.58	27900.55 $\pm$ 37444.32	59100.7 $\pm$ 116454.43	21460.6 $\pm$ 28627.17
	Orange	810694.9 $\pm$ 309662.97	760731.15 $\pm$ 417300.06	755282.7 $\pm$ 347885.64	432110.0 $\pm$ 344369.71	701484.15 $\pm$ 471169.69
	Yellow	46135.9 $\pm$ 39305.67	113784.2 $\pm$ 106383.44	153857.45 $\pm$ 143448.60	116616.1 $\pm$ 94995.56	116365.6 $\pm$ 87574.23
	Cyan	236366.4 $\pm$ 216391.64	171465.9 $\pm$ 177496.27	225215.25 $\pm$ 252531.99	549679.75 $\pm$ 454631.02	370348.25 $\pm$ 377877.27
	Violet	24026.5 $\pm$ 59296.09	10480.5 $\pm$ 12995.54	8770.5 $\pm$ 12240.22	39917.55 $\pm$ 101179.88	6800.7 $\pm$ 8361.73
Saturation	Min	0 $\pm$ 0	0.25 $\pm$ 1.08	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
	Max	250.55 $\pm$ 13.46	255.0 $\pm$ 0	255.0 $\pm$ 0	254.0 $\pm$ 3.72	255.0 $\pm$ 0
	Mean	84.53 $\pm$ 36.30	79.41 $\pm$ 44.6	80.00 $\pm$ 27.95	75.25 $\pm$ 24.85	80.23 $\pm$ 28.83
	Std	51.12 $\pm$ 14.67	48.69 $\pm$ 11.40	57.97 $\pm$ 13.51	47.95 $\pm$ 11.26	43.00 $\pm$ 13.74
	Median	78.75 $\pm$ 47.00	71.2 $\pm$ 52.54	74.1 $\pm$ 38.07	65.3 $\pm$ 29.38	74.5 $\pm$ 34.71
Value	Min	2.75 $\pm$ 5.17	1.45 $\pm$ 3.15	4.8 $\pm$ 10.74	3.0 $\pm$ 7.26	2.2 $\pm$ 4.57
	Max	255.0 $\pm$ 0	255.0 $\pm$ 0	255.0 $\pm$ 0	255.0 $\pm$ 0	254.45 $\pm$ 1.53
	Mean	143.0 $\pm$ 21.01	144.62 $\pm$ 24.42	132.99 $\pm$ 40.71	155.97 $\pm$ 17.01	133.04 $\pm$ 32.27
	Std	63.84 $\pm$ 9.61	61.51 $\pm$ 11.25	59.65 $\pm$ 13.34	59.87 $\pm$ 13.33	58.47 $\pm$ 12.41
	Median	152.6 $\pm$ 29.37	150.45 $\pm$ 34.43	134.4 $\pm$ 53.21	164.9 $\pm$ 26.91	138.1 $\pm$ 47.29

Table 3 Instagram top 5 image labels for 5 sample users

User 1		User 3		User 8		User 12		User 13	
Extraversion		Agreeableness		Conscientiousness		Neuroticism		Openness	
Label	Frequency	Label	Frequency	Label	Frequency	Label	Frequency	Label	Frequency
espres- so	11	lakeside	7	espresso	7	espres- so	6	lake- side	5
plate	6	swan	2	eggnog	6	palace	4	steel_ arch_ bridge	4
cup	6	valley	2	cup	5	mon- astery	4	valley	3
tray	5	flower	2	dog	4	castle	3	palace	3
cof- fee_ mug	5	cup	2	bakery	3	church	3	cross- word_ puzzle	3

used to analyze the content of the images. Table 3 demonstrates the top 5 image labels extracted from the 20 images of the same 5 sample users. These recurring visuals, such as food, landscapes, or objects of daily use, are examined with personality traits. For example, a frequent occurrence of nature- or architecture-related tags may suggest tendencies linked to agreeableness or openness.

Image Textures.

Image textures provide insights into visual preferences, which are influenced by personality traits [33]. This work utilizes the Gray Level Co-Occurrence Matrix (GLCM) to quantify image textural features. GLCM is a statistical method that examines the spatial

Table 4 Instagram image textural features (Mean $\pm$ Std) for 5 sample users

Image Feature	User 1	User 3	User 8	User 12	User 13
	Extraversion	Agreeableness	Conscientiousness	Neuroticism	Openness
Contrast	52.20 $\pm$ 50.68	91.62 $\pm$ 82.68	84.23 $\pm$ 72.69	194.61 $\pm$ 185.64	90.78 $\pm$ 76.7
Dissimilarity	3.16 $\pm$ 1.53	4.03 $\pm$ 2.03	4.02 $\pm$ 2.21	6.22 $\pm$ 3.73	4.33 $\pm$ 2.34
Homogeneity	0.46 $\pm$ 0.12	0.44 $\pm$ 0.11	0.42 $\pm$ 0.13	0.388 $\pm$ 0.13	0.42 $\pm$ 0.13
Energy	0.03 $\pm$ 0.01	0.03 $\pm$ 0.0091	0.03 $\pm$ 0.020	0.036 $\pm$ 0.019	0.04 $\pm$ 0.02
Correlation	0.99 $\pm$ 0.0043	0.98 $\pm$ 0.014	0.988 $\pm$ 0.01	0.97 $\pm$ 0.025	0.98 $\pm$ 0.01
ASM	0.0014 $\pm$ 0.0011	0.0012 $\pm$ 0.0006	0.0018 $\pm$ 0.0023	0.0017 $\pm$ 0.0023	0.0022 $\pm$ 0.0029

Table 5 Instagram features datasets overview

Dataset	Data Type	Dimensionality	Dataset Size
Profile Metrics	Numeric	11	30
Image HSV	Numeric	23	600
Image Labels	Numeric	26	600
Image Texture	Numeric	12	600

relationship between pixel intensities to quantify features such as contrast, dissimilarity, homogeneity, energy, correlation, and Angular Second Moment (ASM). These textural features are retrieved for each liked image.

Contrast measures the intensity difference across the image, reflecting more textured or detailed areas. Similar to contrast, dissimilarity highlights variations in texture without overly penalizing small differences. Furthermore, correlation measures the relationship between pixel intensity distribution. Such measures may indicate a preference for detailed and intricate visuals, which is associated with detail-oriented traits like conscientiousness. On the other hand, homogeneity quantifies the uniformity or consistency of textures. In addition, high energy suggests repetitive patterns or uniform textures. Moreover, Angular Second Moment (ASM) measures texture uniformity, where a high ASM value corresponds to smooth and highly predictable texture. Such textural features suggest a preference for order and simplicity, often linked to traits like agreeableness or calmness. On the other hand, tendencies toward complexity, structure, and harmony are often associated with conscientiousness or agreeableness.

These textural features are computed and aggregated as the mean and standard deviation of the metrics for each user's images. Table 4 depicts the aggregated image textures for the same user samples.

Finally, the extracted liked image features are preprocessed; the metrics features are normalized, and the images are resized before being passed to the ML models. Table 5 provides an overview of the four Instagram datasets.

3.1.3 Ground truth data acquisition

We collect two ground truth datasets for annotating the datasets for personality prediction, as well as validating the accuracy of the career recommendations. The ground truth data were collected from a sample student group through two questionnaires: the Big Five test and the Electronic Emirati Scale for Professional Inclinations (EESPI) test.

In this work, we utilize the Five-Factor Model (FFM), known as the Big Five model, for implicit personality acquisition. The FFM is a widely recognized and extensively validated framework in personality research, with broad applications in areas such as personality-based recommendation systems [33]. This is due to the ease of assessment

through standardized and widely available questionnaires as well as its validity across cultures, demographics, and various contexts [7, 29, 32].

The model's five dimensions are broad enough to capture major variations in personality yet concise, to ensure practical usability. The Big Five model consists of five aspects, namely, extraversion, agreeableness, conscientiousness, neuroticism, and openness [15]. First, extraversion identifies individuals' behavior in social environments. Second, agreeableness addresses the individual's social behavior and interactions with others. It includes aspects such as compassion, politeness, and trust. Third, conscientiousness addresses the ability of a person to organize and complete tasks to accomplish long-term goals. It includes aspects such as orderliness, self-discipline, and reliability. Fourth, neuroticism is concerned with the extent to which a person is prone to negative emotions and mood changes. It includes aspects such as anxiety, depression, and emotional instability. Lastly, openness describes a person's intellectual, artistic, and experiential capabilities. It includes facets such as imagination, aesthetic sensitivity, and intellect.

One of the most common Big Five personality questionnaires is IPIP NEO-PI-R. In this work, we utilize the 120-item IPIP NEO-PI-R [16] version. This personality measurement tool is a multilingual, free, and open-source personality test that can be answered within 10 to 20 min. The questionnaire consists of 120 short questions with a five-scale answer that ranges from very inaccurate to very accurate. The test provides a 0 to 120 score for each of the personality traits. Scores that are above 70 indicate that the individual is high on this trait and vice versa. The results of this test are the scores of each personality trait. The personality with the highest trait is used as a label for annotation. Table 6 demonstrates a sample of Big Five personality test questions.

Along with the Big Five test, the students are asked to answer the Electronic Emirati Scale for Professional Inclinations (EESPI) test. EESPI is a psychometric assessment tool developed to evaluate individuals' interests and professional tendencies, particularly within the cultural context of the United Arab Emirates (UAE). It aims to assist students and professionals in identifying career paths that align with their personal inclinations.

This questionnaire consists of 84 yes/no questions and is available for high school students from grade 9 to grade 12 [34]. This questionnaire guides them to study majors that match their personal preferences by providing scores for each study major. At the end of the test, the student gets the top 3 professional inclinations and study majors. The results of this test are used to validate the proposed recommender system results. Table 7 demonstrates a sample of the EESPI test questions.

Table 6 Sample big five personality test questions

Big Five Personality Test Question

1. Worry about things				
Very Inaccurate	Moderately Inaccurate	Neither Accurate nor Inaccurate	Moderately Accurate	Very Accurate
2. Make friends easily				
Very Inaccurate	Moderately Inaccurate	Neither Accurate nor Inaccurate	Moderately Accurate	Very Accurate
3. Have a vivid imagination				
Very Inaccurate	Moderately Inaccurate	Neither Accurate nor Inaccurate	Moderately Accurate	Very Accurate
4. Trust others				
Very Inaccurate	Moderately Inaccurate	Neither Accurate nor Inaccurate	Moderately Accurate	Very Accurate
5. Complete tasks successfully				
Very Inaccurate	Moderately Inaccurate	Neither Accurate nor Inaccurate	Moderately Accurate	Very Accurate

3.1.4 Ethical approval and consent

The Institutional Review Board (IRB) of the College of Engineering authorized this study, which was carried out in compliance with Abu Dhabi University's ethical guidelines (File Number: CoE-22-05-00017, approval date: May 16, 2022). The Declaration of Helsinki and its subsequent amendments, as well as institutional norms, were adhered to in every procedure.

All participants gave their informed consent before any data was collected, and participation was entirely voluntary. Every participant was older than 18, and the information gathered was anonymized to protect confidentiality and privacy. Instagram data was only accessed with subjects' express consent.

Data collection was limited to content available at the time of consent. Private or inaccessible content was not accessed, and deleted posts were not retrieved. Only anonymized, derived features were used in the analysis, and raw Instagram data were not shared publicly due to ethical and privacy constraints.

3.2 Module 2: personality prediction

The following subsections detail the processes of feature selection, model training, and recommender systems.

3.2.1 Choosing the best feature set for personality prediction through correlation analysis

Pearson correlation is a statistic and inferential measure that indicates the strength and direction of the linear relationship between two variables [81]. Equation 1 shows the Pearson correlation coefficient calculation.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}} \quad (1)$$

where,

r: the Pearson correlation Coefficient.

n: the total number of values.

x: values of the first variable; the Instagram features data.

y: values of the second variable; the Big Five data.

In this work, we apply Pearson correlation between the Big Five personality traits and all Instagram features. The resulting correlation coefficients indicate the dependency between the variables for feature selection. These coefficients identify which Instagram features exhibit strong associations with each of the Big Five personality traits. Features with both high positive and negative correlations are considered as they can significantly impact model performance.

Table 7 Sample EESPI test questions

EESPI test question	Answer	
1. I prefer to work at discovering scientific facts	Yes	No
2. I plan to study a major in engineering	Yes	No
3. I would love to work in robotics	Yes	No
4. I care about archaeological sites	Yes	No
5. I would like to join a medical specialty	Yes	No

Pearson correlation was selected for feature selection due to its suitability for small datasets, computational efficiency, and ease of interpretation. It makes it possible to clearly identify important features by offering insights into the direction and strength of linear correlations between personality traits and Instagram-derived features. In personality-driven educational applications, where comprehending feature-trait correlations is just as crucial as prediction performance, this interpretability is especially crucial.

All extracted Instagram features were first taken into consideration for correlation analysis against the Big Five personality traits in order to guarantee transparency in the feature selection process. The complete feature set comprised picture-based HSV color features, semantic image labels, image texture features derived from GLCM statistics, and profile metrics (followers, followings, posts, likes, and comments). Pearson correlation analysis between each characteristic and the associated personality trait scores was used to choose features. For modeling, features with an absolute correlation coefficient of $|r| > 0.30$ and a statistically significant p-value ($p < 0.05$) were kept. This criterion led to the selection of HSV color features and Instagram profile metrics as inputs to the machine learning models since they showed the strongest and most consistent relationships with personality traits. The final prediction models did not include image label and texture variables that had weaker or inconsistent relationships.

3.2.2 Training machine learning models for personality prediction

For personality prediction, we leverage feature annotated with the Big Five personality traits. To enhance model performance, we select and utilize only the subset of features that exhibit the highest correlations with each of the Big Five traits. We evaluate the predictive capabilities of four Machine Learning (ML) models: Logistic Regression (LR), Linear Discriminant Analysis (LDA), Gaussian Naïve Bayes (GNB), and Support Vector Machine (SVM). The hyperparameters of these models are set to the defaults. These default settings correspond to the standard configurations provided by the scikit-learn library implementations of Logistic Regression, Linear Discriminant Analysis, Gaussian Naïve Bayes, and Support Vector Machine. Using default configurations ensured reproducibility and reduced the risk of overfitting, which is particularly important given the limited dataset size. Model performance is assessed using accuracy, precision, recall, F1-score, and mean AUC as evaluation metrics.

The fundamental mathematical underpinnings of the four machine learning models are briefly outlined below to give a clearer understanding of the predictive techniques employed in this study.

Logistic Regression (LR).

Logistic Regression predicts the probability that a given input belongs to a particular personality class. The model applies a sigmoid transformation to a linear combination of input features:

$$P(y = k|x) = \frac{1}{1 + e^{-(w_k^T x + b_k)}}$$

where w_k and b_k denote the parameters learned for each personality trait k .

Linear Discriminant Analysis (LDA).

LDA assumes that each class follows a Gaussian distribution with a shared covariance matrix. The decision function is:

$$\delta_k(x) = x^T \sum -1 \mu_k^T \sum -1 \mu_k + \ln(\pi_k)$$

where μ_k is the mean of class k , Σ is the common covariance matrix, and π_k is the prior probability.

Gaussian Naïve Bayes (GNB).

GNB predicts personality traits by assuming feature independence and modeling each feature with a Gaussian distribution:

$$P(y = k|x) \propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma_{ki}^2} \exp\left(-\frac{(x_i - \mu_{ki})^2}{2\sigma_{ki}^2}\right) \pi_k$$

where μ_{ki} and σ_{ki}^2 are the mean and variance of feature i for class k .

Support Vector Machine (SVM).

SVM identifies the optimal separating hyperplane by maximizing the margin between classes. The decision boundary is defined as:

$$f(x) = \text{sign}(w^T x + b)$$

where w and b are optimized to maximize the margin $\frac{2}{\|w\|}$. A kernel function can map data to a higher-dimensional space when classes are not linearly separable.

These four models were chosen as baseline classifiers that are easy to understand and appropriate for small datasets. They preserve low computing complexity and robust generalization behavior while representing complementary modeling paradigms probabilistic (LR, GNB), generative (LDA), and margin-based (SVM). While recent research investigates deep and multimodal architectures for personality prediction, traditional machine learning models such as these remain valuable baselines [13, 14, 35]. Deep learning models are less appropriate as first-line methods in small-sample educational studies due to their complexity and data requirements, even if they are capable of capturing intricate non-linear patterns across modalities.

Based on the results of the personality questionnaire, personality labels were used to annotate the users' datasets with personality class labels. In particular, each participant's highest score on the IPIP NEO-PI-R questionnaire was used to establish the dominant Big Five personality feature, which was then used as the class designation.

3.3 Module 3: personalized career recommendation

Recommender systems deploy varying algorithms, including collaborative filtering, content-based filtering, knowledge-based systems, stereotype systems, as well as hybrid systems [36, 37]. This work utilizes a stereotype-based recommender system for career recommendation. Stereotype-based modeling is premised on the assumption that users or items with similar features would share similar broad-level similarities. As such, stereotypes can be viewed as an aggregate set of characteristics of users or items, which enable a recommender system to allocate new users or items to an existing group [38]. Stereotypes might not be an actual representation of a given user group. Rather, stereotypes provide an approximate number of characteristics that are common to a construct (user or item) of interest.

Based on two widely accepted assessments [39, 40], Table 8 summarizes the best academic majors that fit with each personality trait of the Big Five model. This data is used to build the personalized recommender system.

The ML models predict the highest trait, which is then passed to the recommender engine. The recommendation system then provides personalized career recommendations that are the best fit for the students based on their personality.

4 Experimental results

This study involved 30 high school students from the United Arab Emirates. Participants completed the Big Five personality test, followed by the Electronic Emirati Scale for Professional Inclinations (EESPI) test. Subsequently, informed consent was obtained to collect data from their Instagram profiles. This section presents the experimental results of the proposed personalized career recommendation system.

4.1 Correlation analysis for feature selection in personality prediction

Pearson correlation is calculated between the Big Five personality traits and all Instagram features. The p-values and Pearson coefficients (r) are assessed to identify which Instagram features are significant, exhibiting strong associations with each of the Big Five personality traits. Figures 3, 4, 5 and 6 illustrate the correlation heatmaps between the various Instagram feature datasets and the Big Five personality traits.

4.1.1 Correlation between profile metrics and big five

Figure 3 depicts the correlation heatmap, showing the relationships between Instagram profile metrics and the Big Five personality traits using Pearson correlation coefficients (r). As illustrated, extraversion shows the strongest positive correlations with followers ($r = 0.45$), comments ($r = 0.41$), likes ($r = 0.38$), and post counts. This indicates that extraverted individuals are more active, posting and interacting frequently, attracting more attention and feedback. In addition, openness shows moderate positive correlations with followers ($r = 0.28$) and post count ($r = 0.32$). This indicates that open individuals might be more expressive or creative, attracting attention through diverse or interesting content. Meanwhile, agreeableness and conscientiousness show low to moderate correlations, indicating that user activity and visibility metrics aren't strongly impacted. On the other hand, neuroticism shows consistently negative correlations with likes ($r = -0.30$), comments ($r = -0.28$), and followers count ($r = -0.24$). This indicates that users may either avoid high engagement or be less successful in generating interaction, possibly due to anxiety, self-consciousness, or inconsistent behavior. These findings are consistent with the samples demonstrated in Table 1.

Table 8 Recommended study majors/ careers based on the big five personality traits

Big five personality trait	Recommended study major/ career
Extraversion	Sales, public relations, event planning, politics, television (hosting, news-casting, or acting), or education.
Agreeableness	Nurses, counselors, religious leaders, teachers, and founders or members of non-profit organizations or charities.
Conscientiousness	Freelance writers, doctors, actors, entrepreneurs, politicians, or consultants.
Neuroticism	Librarians, yoga instructors, massage therapists, artists, freelance designers, food scientists, or freelance writers.
Openness	Artist, pilot, philosopher, traveler publicist, lawyer, or entrepreneur.



Fig. 3 Instagram profile metrics correlation with the Big Five personality traits

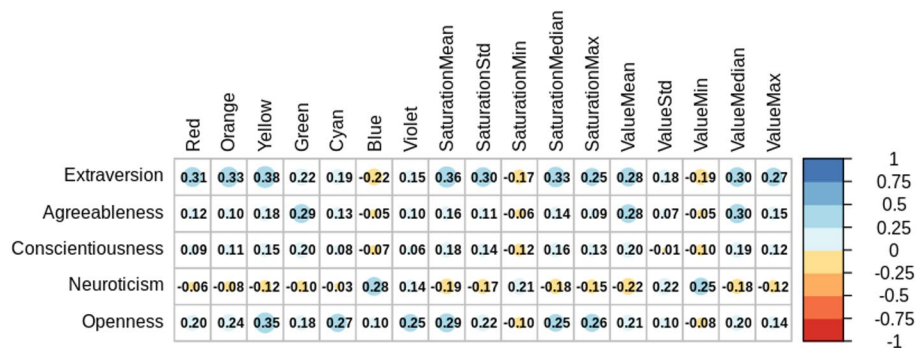


Fig. 4 Instagram image HSV features correlation with the Big Five personality traits

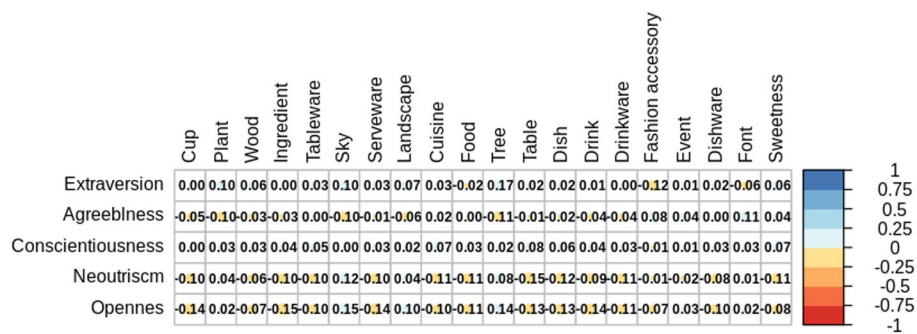


Fig. 5 Top 20 Instagram image labels correlation with the Big Five personality traits

4.1.2 Correlation between image HSV features and big five

Figure 4 depicts the correlation heatmap, showing the relationships between image HSV (Hue, Saturation, Value) features and the Big Five personality traits. This correlation provides nuanced insights into the visual preferences influenced by personality traits. As demonstrated, extraversion shows the strongest positive correlations with green ($r=0.38$), yellow ($r=0.38$), and orange ($r=0.33$). Extraversion is also positively correlated with red, saturation mean, and value mean. This indicates that extroverts tend to share brighter, more colorful images with warm and vivid hues. This aligns with their



Fig. 6 Instagram image texture features correlation with the Big Five personality traits

expressive, energetic, and social nature. Openness shows the second strongest correlation with HSV features. It shows consistent positive correlation with yellow ($r=0.35$), green ($r=0.35$), and orange ($r=0.34$). This indicates that open individuals tend to use a diverse and rich color palette. Their visuals are imaginative and emotionally expressive, reflecting creativity and aesthetic awareness. Agreeableness shows moderate positive correlation with green ($r=0.29$) and value mean ($r=0.30$). This indicates that agreeable individuals prefer friendly, balanced visuals related to natural and soothing colors like green. Conscientious shows weaker overall correlations with green ($r=0.20$) and saturation median ($r=0.19$). This demonstrates that conscientious users don't show strong color preferences, possibly because they focus on consistency or minimalism in their visual choices. On the contrary, neuroticism shows consistent negative correlations with the majority of the HSV features except blue ($r=0.28$). This indicates that neurotic individuals may prefer cooler, darker, or more subdued visuals—often reflecting emotional sensitivity or avoidance of stimulating colors. These findings are consistent with the sample demonstrated in Fig. 2; Table 2.

4.1.3 Correlation between image label and big five

Figure 5 depicts the correlation heatmap, showing the relationships between image labels and the Big Five personality traits. Image labels are descriptive features that reflect underlying personality traits. Hence, the analysis focuses on the top 20 most frequent labels. Extraversion shows weak but consistent positive correlation with “drinkware” ($r=0.12$), “plant” ($r=0.06$), and “wood” ($r=0.06$). This indicates that extroverted users might be more likely to include social or lifestyle elements like drinks or natural settings in their posts, which are associated with social gatherings. Similarly, conscientiousness shows weak positive correlation with “food” ($r=0.08$), “tree” ($r=0.08$), and “dish” ($r=0.06$). This indicates that conscientious users might be more intentional in sharing

lifestyle elements. Openness, on the other hand, shows negative correlation with “cup” ($r = -0.14$) and “tableware” ($r = -0.15$), while showing positive correlation with nature-related elements such as “sky” ($r = 0.15$), “landscape” ($r = 0.12$), and “tree” ($r = 0.13$). This indicates their preference towards outdoor elements, reflecting curiosity, creativity, or a desire for aesthetic richness. Similarly, agreeableness shows positive but weak correlation with “plant” ($r = 0.10$) and “landscape” ($r = 0.06$). This indicates that agreeable individuals may subtly lean toward harmonious, nature-aligned imagery but without strong preferences. Finally, neuroticism shows slight negative correlation with “tree” ($r = -0.15$), “dish” ($r = -0.12$), and “wood” ($r = -0.12$). Users high in neuroticism reflect a tendency to avoid environments linked to calm or stability.

4.1.4 Correlation between image texture and big five

Figure 6 depicts the correlation heatmap, showing the relationships between image texture features and the Big Five personality traits. Texture features, such as contrast, dissimilarity, and homogeneity, capture the structural patterns within posted images. These visual features may indirectly reflect behavioral tendencies, emotional tone, or aesthetic preferences shaped by personality. Extraversion shows weak negative correlation with homogeneity ($r = -0.11$), energy ($r = -0.08$), correlation ($r = -0.09$), and ASM ($r = -0.08$), while displaying slight positive correlation with contrast ($r = 0.08$) and dissimilarity ($r = 0.10$). This pattern may suggest that extroverted individuals prefer images with more visual complexity or variation, consistent with their stimulation-seeking nature. Neuroticism exhibits a positive correlation with contrast ($r = 0.13$) and dissimilarity ($r = 0.12$), and a slight negative correlation with correlation ($r = -0.12$). These findings suggest that users high in neuroticism might favor more visually chaotic or textured imagery, possibly reflecting internal tension or restlessness. Openness is weakly positively associated with contrast ($r = 0.09$) and dissimilarity ($r = 0.10$), and shows negative correlations with homogeneity ($r = -0.08$) and correlation ($r = -0.07$). This indicates a preference for diverse, less uniform visual structures, which aligns with openness-related traits such as creativity and preference for novelty. Agreeableness shows very weak but consistent positive correlations with energy ($r = 0.11$), correlation ($r = 0.07$), and ASM ($r = 0.10$). These subtle patterns may reflect a mild preference for calmer, more uniform visual textures, aligning with agreeable individuals' tendency toward harmony and balance. Conscientiousness, in contrast, shows no strong associations, with all correlation values hovering near zero. This may suggest that texture features alone are not strong indicators of conscientious users' visual preferences or that their posting behavior is more influenced by content organization or context rather than texture.

The identified correlations provide valuable insights into individual preferences that can inform career recommendations. Overall, extraverted individuals, characterized by a preference for vibrant and dynamic content, are likely to excel in careers requiring high levels of social interaction and communication. Conversely, agreeable individuals, who favor smoother textures and exhibit less attraction to nature-themed content, may thrive in roles where empathy, cooperation, and interpersonal skills are paramount. Conscientiousness does not exhibit strong correlations with Instagram features, suggesting a neutral influence on visual preferences; however, individuals with this trait may perform well in environments that prioritize organization, attention to detail, and goal orientation. Neurotic individuals tend to prefer less vivid images and show a slight positive

correlation with nature-related content, indicating potential fulfillment in careers that value creativity, introspection, and attention to detail. Individuals characterized by openness, who engage with both nature-themed and vibrant images, may excel in careers that demand creativity, innovation, and a diverse range of interests. These insights underscore the potential of utilizing social media data to provide personalized career guidance tailored to individual personality traits and visual preferences, thereby offering a novel approach to career counseling and education.

These correlation results are used for feature selection. Both profile metric features and image HSV features show strong correlation with the Big Five personality traits. Hence, the profile metrics and HSV datasets are used to train the predictive models.

4.2 Big five personality prediction using the selected Instagram feature set

After feature selection and dataset construction, four machine learning models—Logistic Regression (LR), Linear Discriminant Analysis (LDA), Gaussian Naive Bayes (GNB), and Support Vector Machine (SVM)—were trained to predict the dominant Big Five personality trait of each student. The classification target was defined as the most dominant trait (i.e., the highest-scoring trait) from the IPIP NEO-PI-R personality assessment. This setup represents a multiclass classification task with five possible class labels: Extraversion, Agreeableness, Conscientiousness, Neuroticism, and Openness.

A stratified hold-out validation approach was used given the small dataset size to assess the model's performance and avoid overfitting. In particular, 17% of the data was set aside for testing and 83% of the data was used for training. This approach was applied consistently across all feature datasets derived from Instagram, including profile metrics and image-based attributes (HSV color features).

Among all feature sets, the Instagram profile metrics dataset yielded the highest predictive performance across models. These features—such as follower count, number of posts, and engagement metrics—were found to be the most discriminative in identifying users' dominant personality traits. Notably, the Logistic Regression (LR) model achieved the best overall results, with an accuracy of 97%, F1-score of 0.98, and a mean AUC of 0.97. This indicates both high precision and strong generalization performance when using behavioral interaction features.

In contrast, models trained on image HSV features showed significantly lower performance. While LDA achieved relatively acceptable results on this dataset (accuracy = 0.82, F1 = 0.90), other models exhibited reduced AUC scores, suggesting that color features alone are less effective for personality classification. The discrepancy highlights the greater predictive value of social interaction patterns over visual aesthetics in modeling personality traits in this context.

Models were evaluated separately on each feature set rather than on combined feature representations, in order to isolate the predictive contribution of each Instagram feature modality. A detailed comparison of performance metrics across models and feature types is presented in Table 9.

4.3 Personalized career recommendations based on big five personality traits

The selected optimal predictive model, Logistic Regression (LR), is employed to predict the students' personality traits based on their Instagram profile metrics. The predicted traits are then passed to the recommender system to generate tailored career

Table 9 Performance of ML predictive models on the profile metrics and HSV features datasets

ML model	Performance Metric	Instagram features datasets	
		Profile metrics	Image HSV
LR	Accuracy	<u>0.97</u>	0.79
	Precision	<u>1.00</u>	1.00
	Recall	<u>0.97</u>	0.97
	F1-score	<u>0.98</u>	0.88
	Mean AUC	<u>0.97</u>	0.49
LDA	Accuracy	0.84	0.82
	Precision	1.00	1.00
	Recall	0.84	0.82
	F1-score	0.91	0.90
	Mean AUC	0.89	0.50
GNB	Accuracy	0.97	0.65
	Precision	1.00	1.00
	Recall	0.97	0.65
	F1-score	0.98	0.78
	Mean AUC	0.90	0.51
SVM	Accuracy	0.91	0.79
	Precision	1.00	1.00
	Recall	0.91	0.79
	F1-score	0.95	0.88
	Mean AUC	0.92	0.43

* Results in **bold** are the highest among the four datasets. The results **underlined** are the highest among the four ML predictive models.

recommendations. For validation, a comparative analysis was conducted between the recommended academic majors generated by the proposed system and the EESPI career preferences. The career recommendation system achieved an accuracy of 90% when compared to the EESPI ground truth. This high level of accuracy underscores the system's effectiveness in aligning predicted personality traits with appropriate career recommendations.

Table 10 presents sample career recommendations from the proposed system based on predicted Big Five personality traits, validated with the ground truth preferences from the EESPI. Notably, there are several instances where the proposed recommendations align closely with the ground truth preferences, as indicated by the matching careers highlighted in bold and underlined.

For instance, User 1 shows high social engagement on Instagram, as demonstrated in Table 1. Therefore, the user's predicted personality is extraversion, which matches expectations. Based on the predicted trait, the system recommends careers in dynamic fields such as sales, public relations, and education, while the ground truth preferences also included applied sciences and social studies. Similarly, User 3, characterized by high agreeableness, had recommended careers in nursing and counseling, which aligns with the ground truth preference for applied sciences and medicine.

According to the EESPI review, the suggested career recommendation system overall achieved a 90% alignment with students' preferences for professional inclinations. Alignment was identified when at least one of the student's stated choice categories overlapped with the suggested career domains, as shown in Table 10.

Random mismatches in which the students' top EESPI preferences did not quite align with the expected personality-based suggestions mostly happened when students stated

Table 10 Sample career recommendations based on the predicted big five personality trait against ground truth

User_ID	Predicted big five personality trait	Proposed system recommendations	Ground truth career preferences
User_1	Extraversion	Sales, public relations, event planning, politics , television (hosting, newscasting, or acting), or education.	Applied sciences, social studies , and politics .
User_3	Agreeableness	Nurses, counselors, religious leaders , teachers, and founders or members of non-profit organizations or charities .	Applied sciences, social studies , and medicine .
User_8	Conscientiousness	Freelance writers, doctors , actors, entrepreneurs, politicians, or consultants .	Applied sciences, social studies , and medicine .
User_12	Neuroticism	Librarians, yoga instructors, massage therapists , artists, freelance, designers, food scientists , or freelance writers	Applied sciences, social studies , and medicine .
User_13	Openness	Pilot, lawyer , philosopher, or entrepreneur	Political, police, and military, as well as law and legal .

* Careers in **bold and underlined** are matching recommendations.

broad or overlapping interests across various fields, underscoring the inherent subjectivity of job preference assessment.

5 Discussion

This study evaluated a machine learning-based framework for predicting personality traits from Instagram activity and generating personalized career recommendations. The analysis addressed three research objectives.

Personality Prediction from Instagram Data (RO1).

The results demonstrate that Instagram data can be used to accurately predict Big Five personality traits. Logistic Regression (LR) trained on Instagram profile metrics achieved the highest classification accuracy of 97%, with a mean AUC of 0.97, precision of 1.00, and F1-score of 0.98. This outperformed Support Vector Machines (accuracy=91%), Gaussian Naive Bayes (accuracy=97%, AUC=0.90), and Linear Discriminant Analysis (accuracy=84%). These results confirm the effectiveness of LR in this domain and indicate that behavioral metadata offers a robust basis for trait prediction. The performance drop observed when using HSV image features (AUC=0.49 with LR) highlights the limited predictive power of low-level visual content alone.

Implications of Feature Correlations for Personality Modeling (RO2).

Correlation analysis between Instagram features and personality scores identified profile metrics as the most informative predictors. Specifically, extraversion exhibited strong positive correlations with followers ($r=0.45$), comments ($r=0.41$), and likes ($r=0.38$). Neuroticism showed negative correlations with likes ($r=-0.30$), comments ($r=-0.28$), and followers ($r=-0.24$). HSV-based visual features showed weaker and more inconsistent associations (e.g., $r=0.38$ for extraversion and orange hue), and texture-based features were marginal (all $|r| < 0.15$). These results suggest that user interaction behavior, rather than content characteristics, provides the most reliable signal for personality inference in adolescent populations.

Validity of Personality-Informed Career Recommendations (RO3).

The recommender system based on predicted personality traits achieved a **90% match rate** with ground truth preferences derived from the Electronic Emirati Scale for Professional Inclinations (EESPI). Sample-level analysis showed alignment between predicted

traits and recommended careers, e.g., students classified as extraverted were matched with roles in education and public relations, consistent with EESPI outputs. These findings indicate that inferred personality traits, when accurately modeled, can serve as valid proxies for explicit preference inputs in career recommendation systems.

The combined goals of interpretability and robustness under small-sample settings informed the modeling decisions used in this study. In order to minimize the risk of overfitting associated with extremely complicated models and to enable unambiguous interpretation of the relationship between Instagram-derived variables and personality traits, classical machine learning models were chosen as transparent baselines. This objective was further aided by feature selection using Pearson correlation, which isolated features with significant linear connections and produced a condensed and informative representation dominated by profile metrics and chosen HSV features. Although the provided results show good discriminative performance, careful interpretation is required due to the small sample size. Therefore, rather than serving as final performance standards, the results should be seen as evidence of viability, emphasizing the potential of behavioral social media elements for personality-aware career advising. More thorough validation techniques and wider generalization across student demographics would be made possible by expanding the dataset.

6 Evaluation considerations and limitations

A stratified hold-out validation approach was used in place of cross-validation due to the pilot study's small sample size. Applying k-fold cross-validation to relatively small datasets might result in significant variance across folds and incorrect estimates when class distributions are sparse, despite the fact that it is frequently employed to get more stable performance estimates [41]. The hold-out split that was selected preserved enough samples for training while offering a transparent and cautious evaluation of the model's generalization on unknown input.

The small number of participants in this study limits the use of nested k-fold cross-validation, which could further minimize model selection bias and variation. Repeated partitioning would significantly lower the effective training data per fold with only 30 instances, possibly hiding rather than revealing significant patterns. Nested cross-validation works well in moderate-to-large datasets with enough samples per fold, according to previous research [42]. More robust validation techniques, such as nested cross-validation, will be possible with future work involving bigger datasets.

Given the magnitude of the dataset, overfitting cannot be completely ruled out, even though high performance metrics particularly the strong AUC values attained using Logistic Regression indicate good class separability. Although performance estimates on short datasets may be optimistic and should be evaluated cautiously, high AUC values indicate strong discriminative power of the chosen features and models [43, 44]. Therefore, rather than being absolute performance assures, these data are best understood as proof of viability. While noting that larger-scale validation is necessary for greater applicability, the findings show that behavioral traits extracted from Instagram carry significant personality cues.

False positives, as seen through the lens of error analysis, are situations in which a personality trait is predicted even though it is not prevalent in the questionnaire findings, which may result in less appropriate career choices. False negatives, on the other hand,

indicate overlooked dominant features, which could lead to an underrepresentation of a student's core inclinations. In educational decision-support systems, where predictive models should support human judgment and counseling processes rather than replace them, the interpretation of false positives and false negatives is especially crucial [45].

7 Conclusions

This study introduced a machine learning framework for inferring Big Five personality traits from Instagram activity and generating personalized career recommendations for high school students. Experimental results demonstrated that personality traits can be predicted with high accuracy using social media profile metrics, with Logistic Regression achieving 97% accuracy and a mean AUC of 0.97. Comparative analysis of feature sets showed that behavioral metadata—such as follower count, post frequency, and engagement levels—outperformed image-based features (e.g., HSV color and texture) in predictive performance, indicating that social interaction patterns are more informative indicators of personality than visual content alone. Moreover, the proposed system achieved a 90% agreement between its recommendations and students' stated preferences, as validated against a standardized career assessment (EESPI), confirming the practical relevance of personality-informed recommendation. By eliminating the need for explicit self-report questionnaires, the framework addresses common limitations in existing systems—namely, respondent bias, incomplete inputs, and low self-awareness in adolescent users. Overall, the findings support the use of passively collected social media data for scalable, low-burden personality modeling and trait-aligned career guidance, contributing to the advancement of data-driven approaches in educational and behavioral computing. This study has limitations pertaining to sample size and cross-sectional data collecting, despite its encouraging findings. In order to better capture personality evolution and enhance suggestion stability, future research will examine bigger and more diverse samples as well as the integration of temporal data such as posting frequency trends, engagement characteristics across time, and continuous changes in visual behavior. The system's capacity to simulate student behavior in a more realistic and context-aware way may be further improved by incorporating time-based signals.

Author contributions

H.I. Conceptualization, investigation, methodology, formal analysis, visualization, user study design, writing original manuscript. M.A. Conceptualization, software, data curation, user study. A.K. Fund acquisition, editing, and reviewing.

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Data availability

The anonymized dataset supporting the findings of this study (including derived features from Instagram metadata and personality trait labels) is available from the corresponding author upon reasonable request. Due to privacy and ethical restrictions, raw Instagram data cannot be shared publicly, but anonymized derived datasets may be provided subject to institutional approval.

Declarations

Ethics approval and consent to participate

This study was approved by the Institutional Review Board of Abu Dhabi University, College of Engineering (File Number: CoE-22-05-00017, approval date: May 16, 2022) and conducted in accordance with institutional guidelines and the 1964 Helsinki Declaration and its later amendments. Informed consent was obtained from all participants (all were above 16 years old). Participation was voluntary, and informed consent was obtained from all participants. All participants were above 16 years old, and their data were anonymized to ensure privacy and confidentiality.

Competing interests

The authors declare no competing interests.

Consent to publish

Not applicable.

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